TITANIC SURVIVAL PREDICTION

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Project Review

Overview

The Titanic Survival Prediction project aims to predict whether a passenger survived the Titanic disaster using Machine Learning techniques. The project uses passenger information such as age, gender, ticket class, fare, and family details to build a predictive model.

Dataset Description

The dataset contains information about Titanic passengers, including demographic details, ticket information, and survival status. The target variable is "Survived," where 1 indicates survival and 0 indicates non-survival.

Data Preprocessing

Several preprocessing steps were performed to prepare the dataset for machine learning:

- Handled missing values in relevant columns.
- Removed unnecessary features with excessive missing data.
- Converted categorical variables into numerical format using encoding techniques.
- Selected important features for model training.

Model Development

A Random Forest Classifier was used to train the prediction model. The dataset was divided into training and testing sets using an 80:20 ratio. The model was trained on the training data and evaluated on the testing data.

Results

The trained model achieved an accuracy of **82%** on the test dataset. This indicates that the model can correctly predict passenger survival for most cases and demonstrates good predictive performance.

Key Findings

- Gender and passenger class were among the most influential factors affecting survival.
- Passengers in higher classes had a greater chance of survival.
- Female passengers generally showed higher survival rates compared to male passengers.
- The Random Forest algorithm provided reliable and accurate predictions.

Conclusion

The Titanic Survival Prediction project successfully demonstrates the application of Machine Learning for classification problems. Through data preprocessing, feature selection, and model training, an accuracy of **82%** was achieved. The project highlights the importance of data analysis and predictive modeling in solving real-world problems and serves as a strong foundation for learning machine learning concepts.